

Shamitha Raj

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EDUCATION

National Institute of Technology Durgapur

B.Tech in Mathematics and Computing | CGPA: 7.57/10, Relevant Coursework: Machine Learning, Probability & Statistics, Linear Algebra, Database Management Systems, Data Structures & Algorithms, Optimization

Expected Graduation: 2028

EXPERIENCE

AI/ML Project Intern

Infosys Springboard Internship

Group Project

- Collaborated on the development of an agentic AI-based research assistant using LangGraph and LLM orchestration.
- Applied concepts in generative AI, retrieval systems, prompt engineering, and modular agent workflows.

PROJECTS

AccuSearch – Production-Ready RAG-Based Study Assistant | Tech Stack: [Python](#), [FastAPI](#), [LangChain](#), [ChromaDB](#), [PostgreSQL](#), [Gemini API](#), [Docker](#), [React](#), [Railway](#), [Vercel](#)

- Built a full-stack Retrieval-Augmented Generation (RAG) system for contextual question answering from academic PDFs using semantic retrieval pipelines.
- Engineered an embedding-based document ingestion pipeline with chunking and vector search using ChromaDB and HuggingFace embeddings.
- Developed an LLM-powered quiz generation module to automatically generate contextual MCQs from uploaded study material.
- Designed a persistent multi-conversation architecture using PostgreSQL and SQLAlchemy ORM for scalable session management.
- Dockerized and deployed the complete application using Railway and Vercel, enabling production-ready cloud deployment.

OpenDeepResearcher – Agentic LLM Research Framework | Tech Stack: [Python](#), [LangGraph](#), [LangChain](#), [Tavily API](#), [Ollama](#), [Streamlit](#), [Git](#), [Qwen LLM](#)

- Built an agentic AI research framework capable of autonomously planning, retrieving, analyzing, and synthesizing information from real-time web sources.
- Designed a multi-agent architecture comprising Planner, Searcher, and Writer agents orchestrated using LangGraph.
- Integrated LLM reasoning pipelines with Tavily web search APIs to generate structured, citation-backed research reports from complex user queries.
- Implemented session memory and conversational continuity to support iterative, multi-step research workflows.
- Developed an interactive ChatGPT-style research interface for structured output visualization.

AccuIQ – Production-Grade Multi-Agent AI Analytics Platform | Tech Stack: Python, FastAPI, PostgreSQL, LangGraph, Docker, ChromaDB, Gemma, Qwen, OCR

- Built an end-to-end AI platform that transforms examination PDFs into structured academic intelligence through a scalable multi-agent pipeline.
- Developed an asynchronous workflow for OCR, question extraction, semantic duplicate detection, metadata generation, and document processing.
- Designed a LangGraph-based orchestration framework coordinating specialized AI agents for document classification, information extraction, validation, and analytics.
- Built an analytics engine to uncover examination trends, topic importance, marks distribution, and recurring question patterns across academic years.

TECHNICAL SKILLS

Programming Languages: Python, SQL

Machine Learning & AI: TensorFlow, Keras, LangChain, LangGraph, RAG Systems, NLP, HuggingFace Embeddings, LLMs

Backend & Databases: FastAPI, PostgreSQL, SQLAlchemy, ChromaDB

Frameworks & Tools: Docker, Git, GitHub, Streamlit, Railway, Vercel, Google Colab

OPEN SOURCE CONTRIBUTIONS:

Contributed to ImprovedTube (750K+ users) through feature implementation and issue resolution.

Contributed to OSV.dev by improving structured vulnerability datasets.

Contributed to MemMachine through feature enhancements and optimisation efforts.

CERTIFICATIONS

FreeCodeCamp – Machine Learning Certification

Infosys – Artificial Intelligence Certification

Infosys – Generative AI Certification

ACHIEVEMENTS:

- Solved 50+ algorithmic problems on LeetCode covering Graphs, Trees, BFS/DFS, Scheduling and Interval-based problems.
- Participated in Flipkart GrID, Code with Cisco, HackOn, Tata Crucible and HP Power Lab Hackathon/Case Competition.
- Associated with HP Power Lab Hackathon/Case Competition